SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1714**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks:30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I M.Gokul 192565067. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

20+28+22+16+13

89

Employee Certification

1) propositional logic.

let:

T= Kevin Completed the training

C= Kevin receives Certification

Rule: $T \rightarrow C$

fact: T

Goal: C

2) CNF and Negated Goal

CNF : $T \rightarrow G$

Negated goal : $\neg T \vee G$

clauses : $- G$

$(\neg T \vee G)$, T, $\neg G$.

3) Resolution:

Resolve: $(\neg T \vee G)$

with: T

Result: C

Resolve : C

with : $\neg C$

$\boxed{\square}$

Result: ... Certification

2.) medical Attention alert

1.) propositional logic

let :

F = Ravi has a fever

M = Ravi needs medical attention.

Rule : $F \to M$

fact : F

2.) CNF $F \to M$

Negated goal: $\neg M$

3.) Resolution :

Resolve $\neg F \lor M$

with : F

Result : M

with : $\neg M$

Result : $\boxed{\nabla}$

Conclusion

$\boxed{\text{Ravi needs medical attention}}$

Library late fine

1) propositional logic

let :

L = Book was returned late

F = Fine is charged.

Rule : $L \rightarrow F$

fact : L

2) CNF :

$L \rightarrow F$

CNF : $\neg L \lor F$

Negated goal : $\neg F$

3) Resolution:

Resolve : $\neg L \lor F$

with : L

Result : F

Resolve : F

with : $\neg F$

Result : $\square$

charged

4) flight Delay notification

let:

D = flight 202 is delayed

N = Passenger are notified.

Rule:

$D \rightarrow N$

fact: D

Goal: N

2) CNF

CNF: $D \rightarrow N$

Negated goal: $\neg D \vee N$

$\neg N$

3) Resolution:

Resolve: $\neg D \vee N$

with: D

Result: N

Resolve: N

with: $\neg N$

Result: $\boxed{\square}$

Conclusion:

$\boxed{\text{passengers are notified}}$

) Employee Benefits Eligibility

1.) propositional logic

let:

P = Employee finished probation

S = Employee becomes permanent staff

B = Employee gets full benefits

Rules:

$P \rightarrow S$

$S \rightarrow B$

fact: P

Goal: B

2.) Convert to CNF

first rule: $P \rightarrow S$

CNF: $\neg P \vee S$

second rule: $S \rightarrow B$

CNF:

fact: P

3.) Negate the Goal

Goal: B

Negated goal: $\neg B$

Complete Clause:

$(\neg P \vee S)$

$(\neg S \vee B)$

P

$\neg B$

4.) Apply Resolution

Step 1:

Resolve: $\neg P \vee S$

with: P

Result: S

Step 2:

Resolve: $\neg S \vee B$

Result: S

B

Step 3:

Resolve: B

with: $\neg B$

Result: $\boxed{\square}$

Therefore:

The employee gets full benefits

$\boxed{B = \text{True}}$